

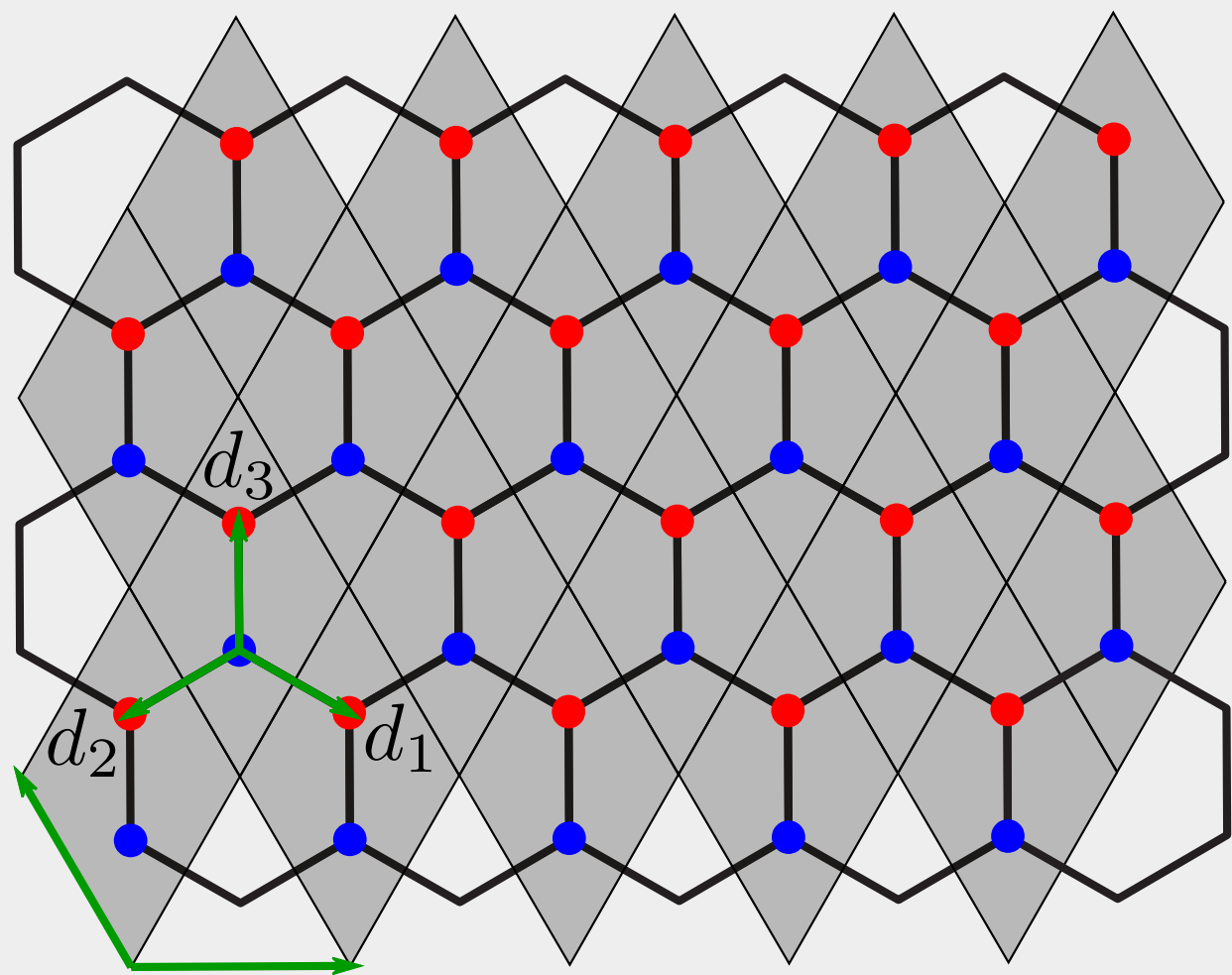
Curvature and Torsion in the Continuum Theory of Lattice Defects

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Offen im Denken

Continuum Theory - Graphene



Tight-binding Hamiltonian

$$H = \sum_{\vec{r}} \sum_{j=1}^3 \left(a_{\vec{r}}^{\dagger} b_{\vec{r}+\vec{d}_j} + b_{\vec{r}+\vec{d}_j}^{\dagger} a_{\vec{r}} \right)$$

Fourier transform

$$i\partial_t \vec{\psi}(\vec{k}) = \mathbf{h}(\vec{k}) \vec{\psi}(\vec{k})$$
$$\mathbf{h}(\vec{k}) = \begin{pmatrix} 0 & f(\vec{k}) \\ f^*(\vec{k}) & 0 \end{pmatrix}, \quad f(\vec{k}) = \sum_{j=1}^3 e^{i\vec{k} \cdot \vec{d}_j}$$

Linear dispersion around Dirac points

$$\vec{k} = \vec{K}^{(s)} + \delta\vec{k} \implies \mathbf{h}(\vec{k}) \approx -\frac{3}{2} \delta\vec{k} \cdot \vec{\sigma}$$

Continuum limit

$$i\partial_t \vec{\phi}(\vec{r}) = i c \vec{\sigma}^i \partial_i \vec{\phi}(\vec{r}), \quad \rho = \vec{\phi}^{\dagger} \vec{\phi}, \quad \vec{j} = -c \vec{\phi}^{\dagger} \vec{\sigma} \vec{\phi}$$
$$\Updownarrow$$
$$i\gamma^{\mu} \partial_{\mu} \phi = 0, \quad \gamma^{\mu} = \sigma^{\mu} \sigma^3, \quad \mu \in \{0, 1, 2\}$$

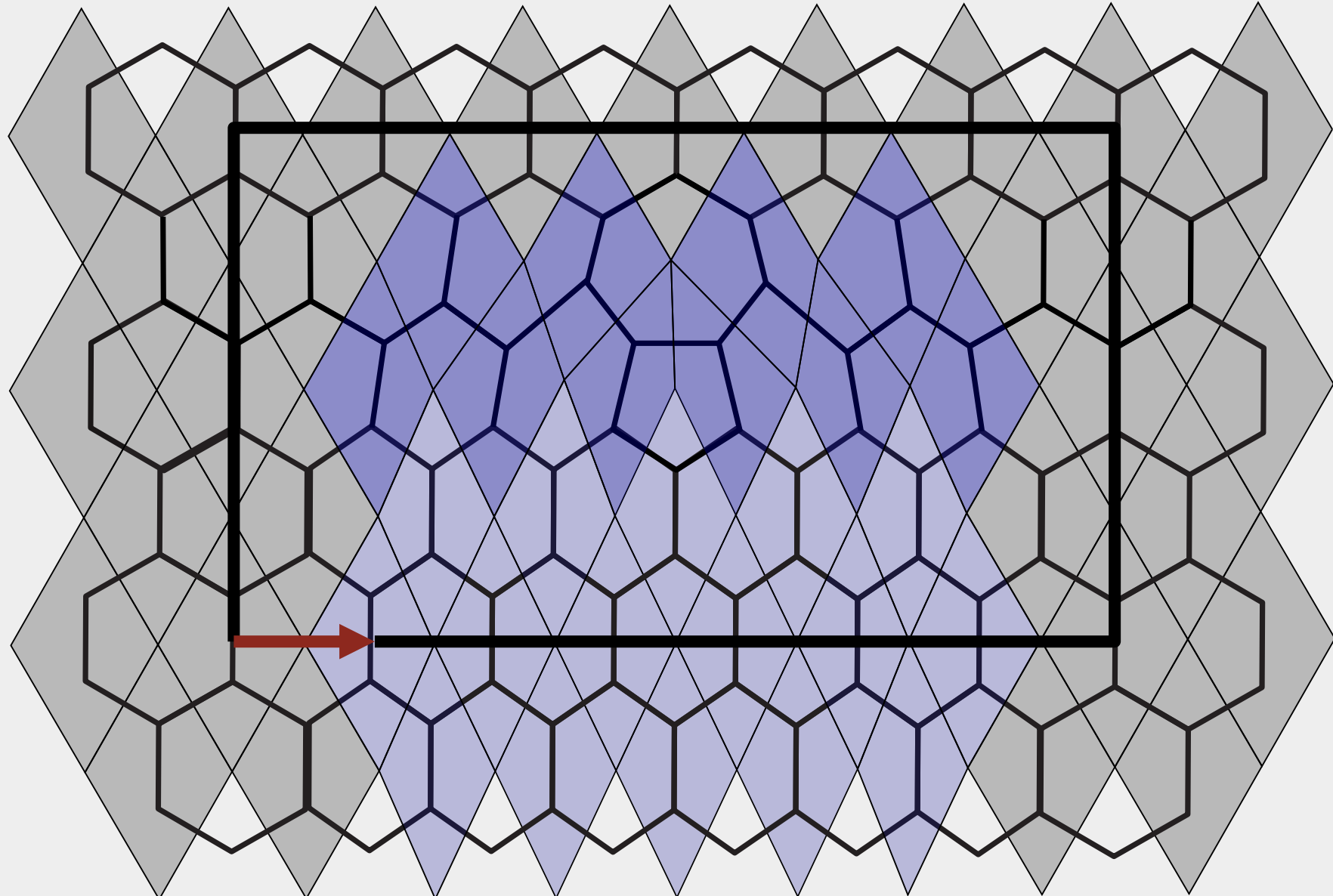
TODO:
Riemannian Geometry basics
- Manifold
- Covariant Derivative

TODO:
Numerical implementation of
the tight binding hamiltonian
- details on defect modeling
- unitary timeevolution

TODO:
GR basics
- Geodesics/Autoparallels
- Curvature
- Torsion

TODO:
Numerical results
(tight binding vs. autoparallels)
Plot Scattering angle over defects

Burgers vector in 5-7 defects



One 5-7 defect causes a lattice dislocation which can be described by the Burgers vector $\vec{b} = \sqrt{3} \vec{u}_x$

TODO:
Torsion <-> metric mapping
- Riemann curvature
- Autoparallels
->Problem with parallel transport
and tangent space in general

TODO:
Outlook
- Dirac equation on curved
backgroun with torsion
- Numeric simulations of this
for some torsion fields

TODO:
References

TODO:
In depth visual explanation of
torsion and how it is analogous
to the burgers vector